

CS230

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Agenda

01 AI and the Age of Acceleration

Diminishing half-life of skills

Enterprises racing towards AI & agent transformation

02 The AI-ready workforce

Productivity enhancements

Future-fit skills

03 The Future of Careers & Organizations

Skills-based organizations

Career tips & tricks



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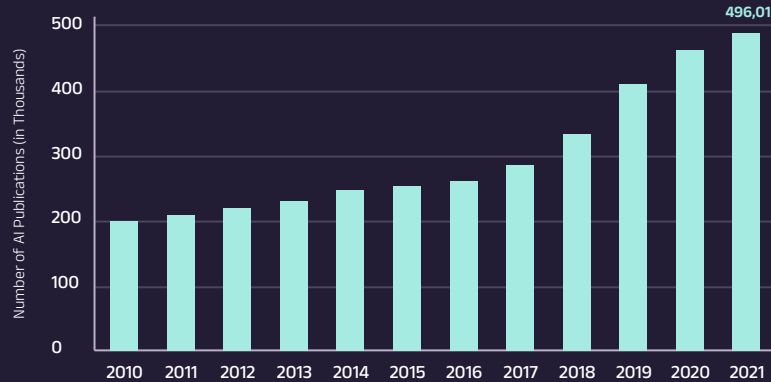
Career tips & tricks



AI research and development is on the rise across the board.

Number of AI Publications in the World, 2010-21

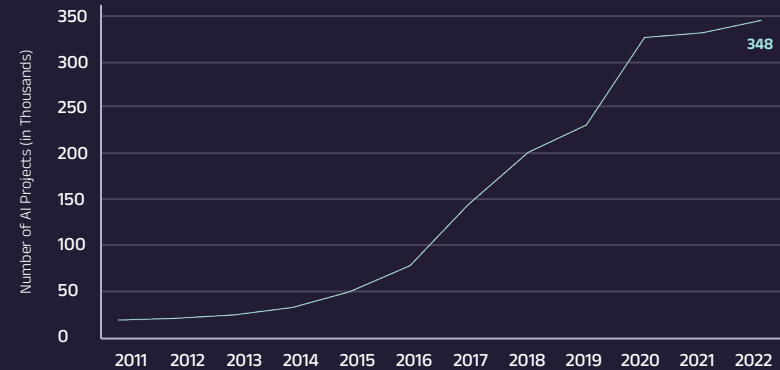
Source: Center for Security and Emerging Technologies, 2022 | Chart: 2023 AI Index Report



From 2010 to 2021, the total number of AI publications more than doubled, growing from 200,000 in 2010 to almost 500,000 in 2021.

Number of GitHub AI Project, 2011-22

Source: GitHub, 2022; OECD.AI, 2022 | Chart: 2023 AI Index Report



Since 2011, the total number of AI-related GitHub projects has steadily increased, growing from 1,536 in 2011 to 347,934 in 2022.



2015

Modern Deep Learning Frameworks

2016

Generative Adversarial Networks

2017

Transformer Architecture

2018

Responsible AI

2019

ML Ops Tools

2020-22

Large Language Models & Latent Diffusion Models

2023

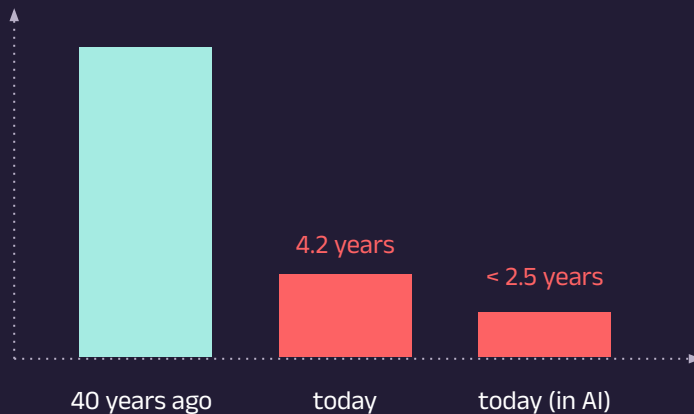
Generative AI & Multi-Modal AI



2024 AI Agents

Rapid adoption of LLMs drives wave of new AI capabilities. Companies are seeing the need to adopt or become displaced.

How long is a skill useful?



The half-life of skills is 80% shorter than 40 years ago.
(More: [World Economic Forum](#))

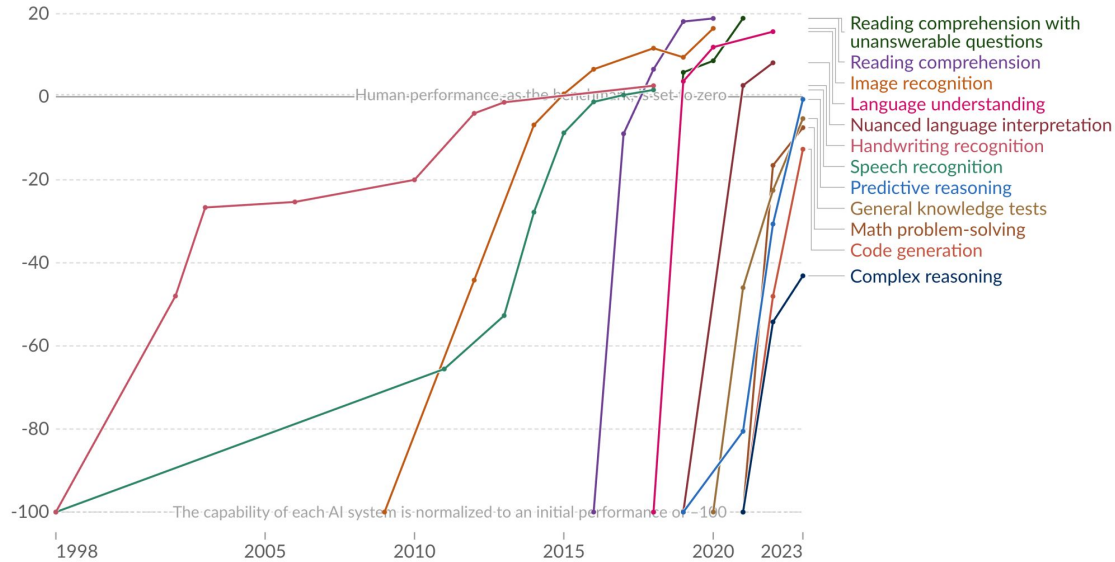


AI is beating human performance in a variety of tasks

Test scores of AI systems on various capabilities relative to human performance

Our World
in Data

Within each domain, the initial performance of the AI is set to -100. Human performance is used as a baseline, set to zero. When the AI's performance crosses the zero line, it scored more points than humans.

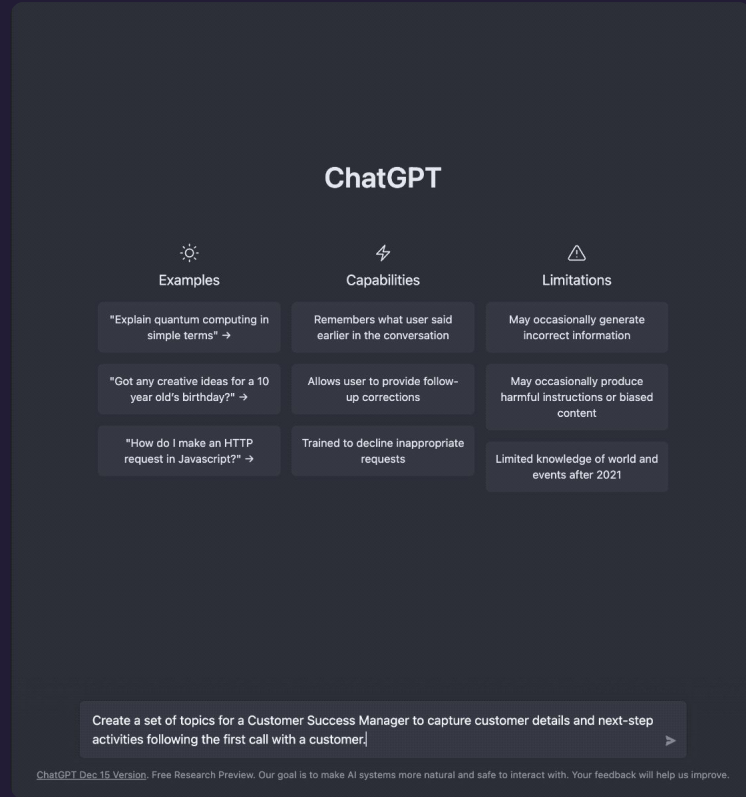
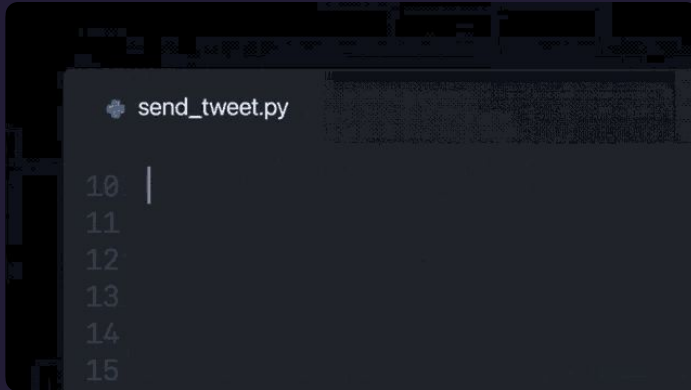


Data source: Kiela et al. (2023)

OurWorldInData.org/artificial-intelligence | CC BY

Note: For each capability, the first year always shows a baseline of -100, even if better performance was recorded later that year.

AI is transforming areas once exclusive to human expertise



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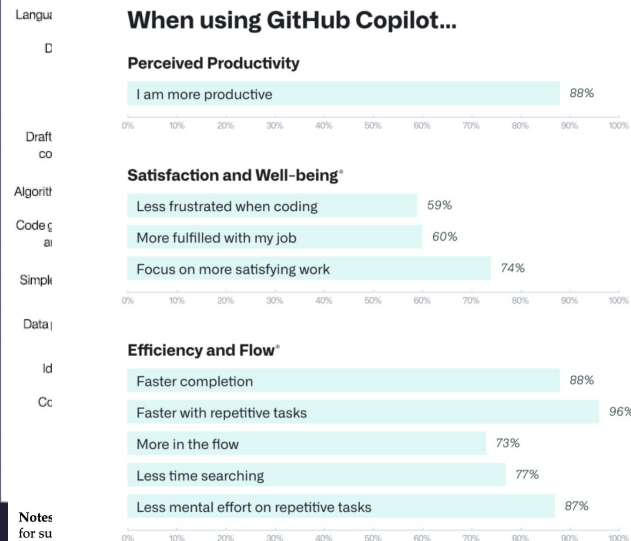
Independent research reveals significant productivity gains attributed to AI

A [study from MIT](#) finds that “AI increased productivity for workers assigned tasks like writing cover letters, delicate emails, and cost-benefit analyses. Access to ChatGPT decreased the time it took workers to complete the tasks by 40%, and output quality, as measured by independent evaluators, rose by 18%

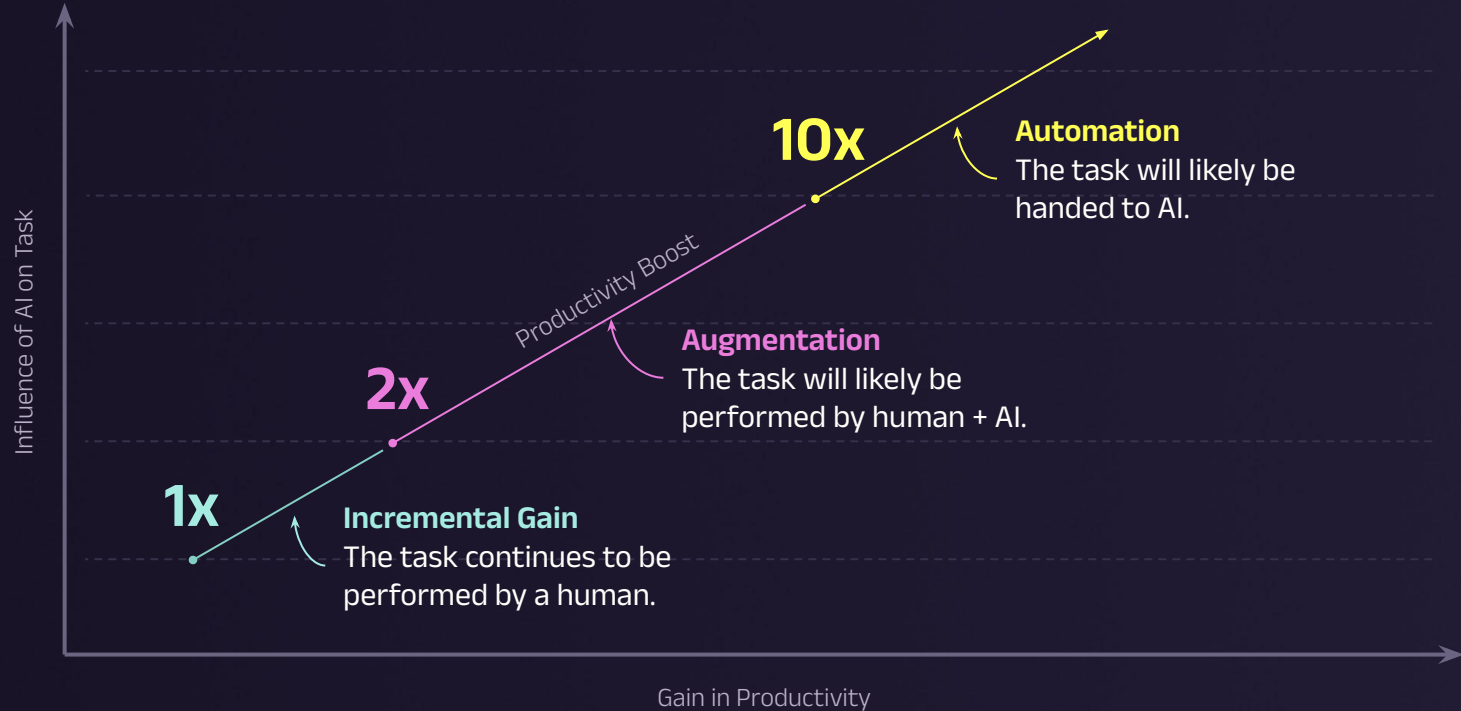
AI tools like Copilot are helping workers. Results of a [GitHub survey](#) on the use of Copilot find that 88% of respondents feel more productive when using the system, 74% feel they are able to focus on more satisfying work, and 88% feel they are able to complete tasks more quickly.

A recent [Workera study](#) reveals that 76% of participants have incorporated Gen AI to enhance their work methods. These individuals note a remarkable 168% productivity boost in basic data and software tasks. Among the 24% not yet utilizing Gen AI, three-quarters anticipate its adoption in their workplace shortly.

Table: Survey responses measuring dimensions of developer productivity when using GitHub Copilot

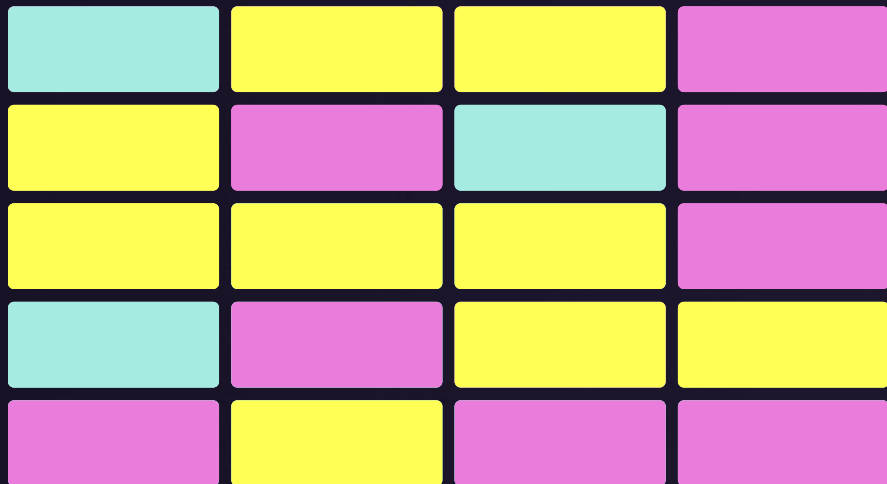


Productivity improvements affect future labor demand.



Jobs consist of tasks that are differentially influenced by AI.

JOB



- Cyan Task is unchanged (productivity gain $< 2x$)
- Pink Task is augmented (productivity gain = 2-9x)
- Yellow Task is automated (productivity gain $> 10x$)

Job is unchanged

Examples: Electricians, Plumbers, etc.

Job is redefined or merged with another job.

Examples: Content Marketer, Creative Designer, etc.

Job is gone

Research from OECD and OpenAI suggests that we will soon see mass job losses in skilled professions, including law, medicine, and finance - OECD warning that 27% of jobs are in "high-risk" professions. McKinsey estimated that with generative AI, 30% of hours worked today could be automated by 2030.

The "Jagged" Frontier (Dell'Acqua et al., 2023): For some tasks, AI exhibits remarkable capabilities, while for others, it may falter either obviously or subtly. Without extensive AI knowledge, distinguishing between the two can be challenging.



Mastering AI skills: A must-have for every professional, technical and beyond

88% Non-technical employees

Examples (non inclusive)

- administrators
- cashiers
- customer care workers
- educators
- healthcare workers
- human resources workers
- managers
- nurses
- retail associates
- salespeople
- supervisors

12% Technical employees

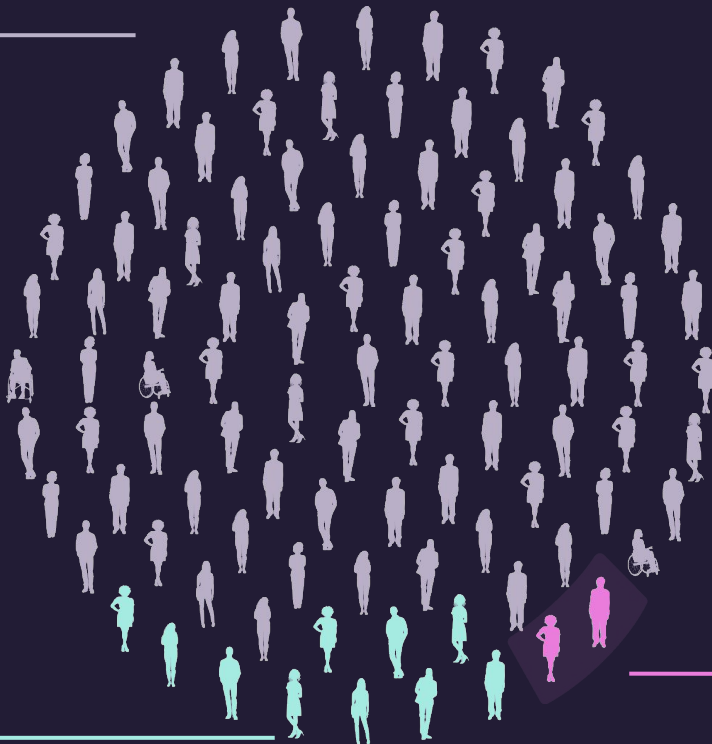
Examples (non inclusive)

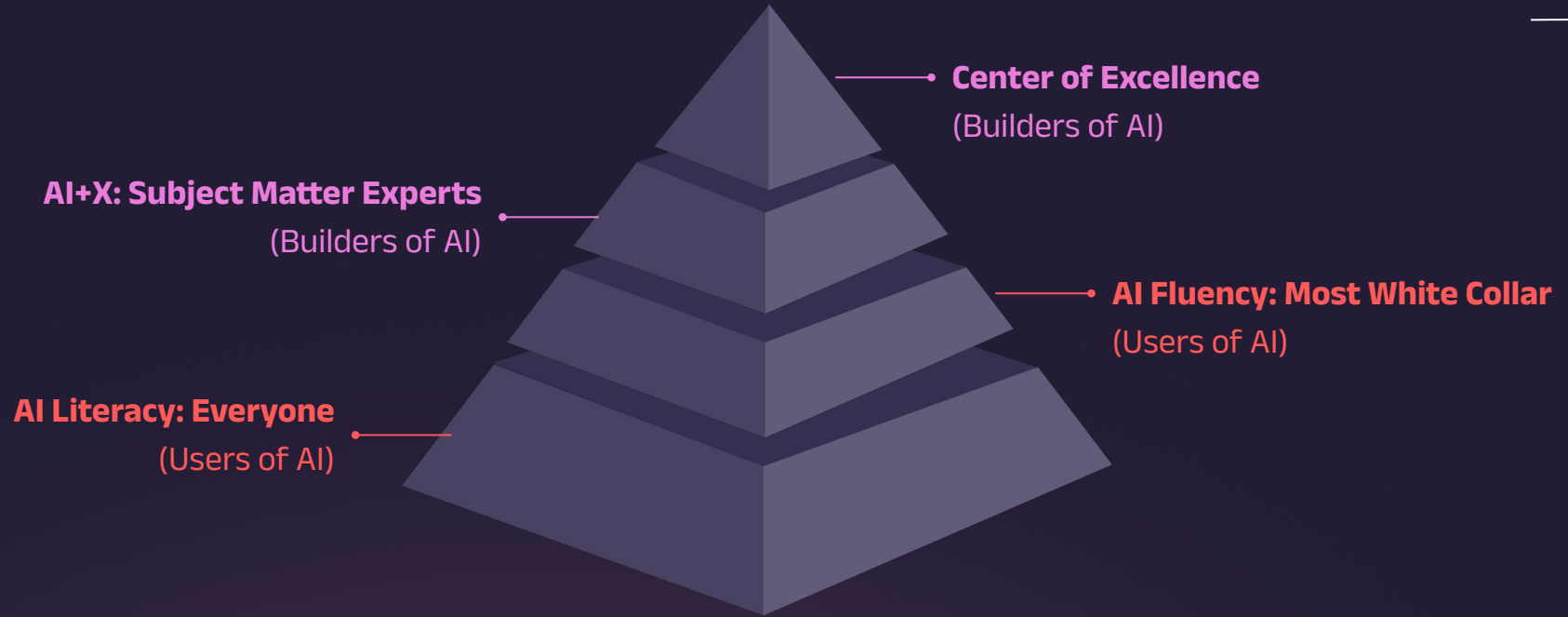
- developers
- engineers
- IT workers
- programmers
- software developers
- software engineers
- system developers
- technologists

2% Technical employees who are generative AI adjacent

Examples (non inclusive)

- data analysts
- data engineers
- data scientists
- machine learning specialists
- researchers
- scientists



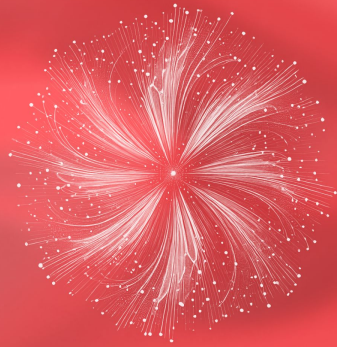


The skills to thrive in the AI world fit in 3 pillars

Let's go to
Mentimeter

AI Foundations

Equip your team with practical AI skills

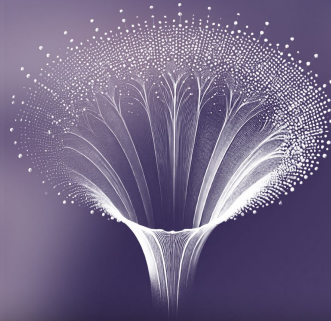


Competencies

- AI Essentials*
- GenAI Essentials*
- Responsible AI

PowerSkills for All

Develop essential competencies for a modern workforce like problem solving, navigating change, and communication.

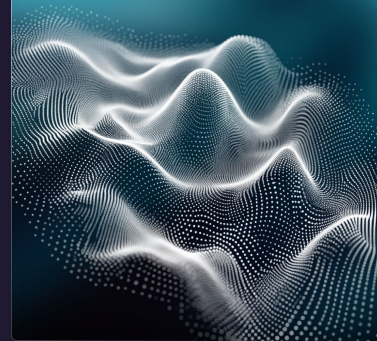


Competencies

- Problem Solving Essentials*
- Effective Learning
- Resilience & Adaptability
- Effective Communication
- Teamwork

Data Literacy

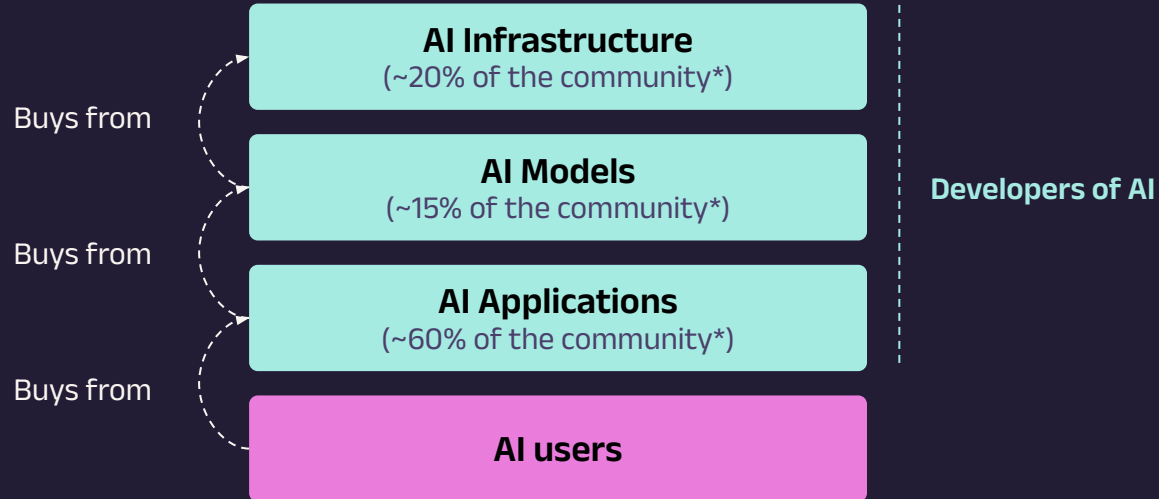
Enhance data proficiency and identify growth opportunities



Competencies

- Data Literacy Essentials*
- Data Storytelling
- Digital Transformation Essentials*

The types of skills needed in the workforce are aligned with AI market segments and their relative sizes.

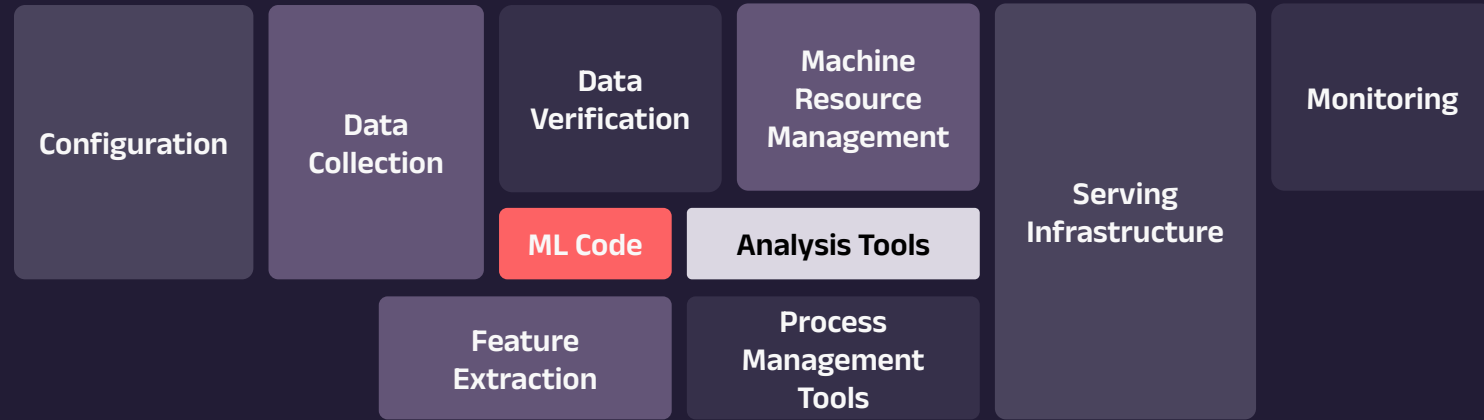


For products in AI infrastructure and AI models layers to make money, you need AI applications to make even more money. The majority of AI developers work in the AI application segment.

*High-level estimates from Workera through survey and analysis (2023)]

*Goldman Sachs Research: AI investment forecast to approach \$200 billion globally by 2025 (2023)]

Deploying AI systems necessitates the coordination of various complex subsystems. Training models represents just a fraction of the overall challenge



AI teams require experts from a wide variety of subject matter expertise. Almost no single person will have all skills.



Trending AI skills (Workera)

Skills for Everyone

AI Foundations

AI Essentials
GenAI Essentials
Responsible AI

Power Skills

Effective Communication
Effective Learning
Problem Solving Essentials
Resilience & Adaptability
Teamwork

Data Foundations

Data Literacy Essentials
Digital Transformation
Data Storytelling Essentials

Future Fit Skills

Data & AI for Business

Data Analytics
AI & Data Communication
Data Decision-Making
Digital Leadership
Foundations
Info. Security Foundations
Marketing Analytics
Data Visualization

Other / Innovation

Product Mgmt
Project Mgmt
UX Fundamentals

Skills for Technical Teams

Software Engineering

Software Quality Assurance
Software Eng. Foundations
Web Apps
Software Architectures
Pair Coding with AI
GenAI For Developers
DevSecOps
Agile with Scrum

Languages (Basic & Advanced)

Java
JavaScript
CSS
HTML
C++
Python
SQL
Pytorch
Tensorflow

Platforms, Libraries, Frameworks

(Basic & Advanced)
GIT
Node.js
Docker
React
Spring Boot
Angular

Cloud & Data Engineering

Cloud Computing*

Data Struct. & Algos
Databases
Distributed Computing
ETL
IT Infrastructure
Databases
Vector Databases
* with AWS, Azure or GCP

Data Science and Analytics

Data Governance
Data Mining
Data Science Processes
GenAI for Data Analytics
Business Analytics
Machine Learning*
Statistical Programming**
*with Tensorflow or PyTorch
** with R, SQL or Python

Other / Innovation

Problem Solving Techniques (Tech)

Operationalizing AI

Deep Learning

MLOps Culture
Responsible AI (Technical)
Building AI Agent Systems
ChatGPT API
LLMOps
Security & Data Gov. in GenAI

Advanced AI Modelling

Computer Vision
NLP
Speech Recognition
Information Retrieval
Large Vision Models
Neural Acceleration Tech
Security in AI
Diffusion Models
Large Language Models
RAG

Cybersecurity

Cybersecurity
Digital Forensics Found
Network Security
Penetration Test. Found

Note: Frequently update



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We are speeding towards a skills-based economy.

73%
of workers

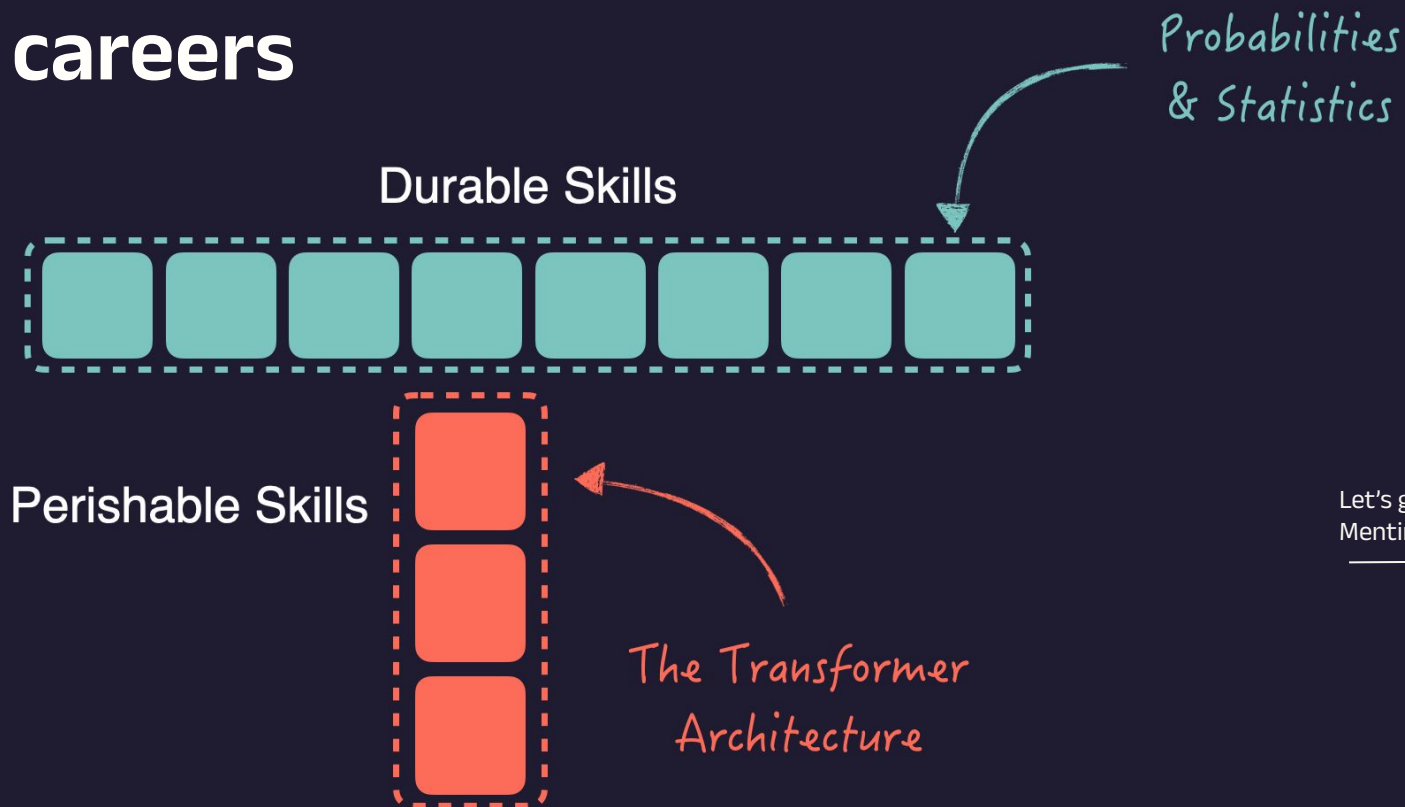
say skills-based practices
would improve their overall
experience at work

89%
of executives

say skills are becoming important
for the way organizations define
work, deploy talent, manage
careers, and value their employees



T-shape careers



learn·ing ve·loc·i·ty

noun

Let's go to
Mentimeter



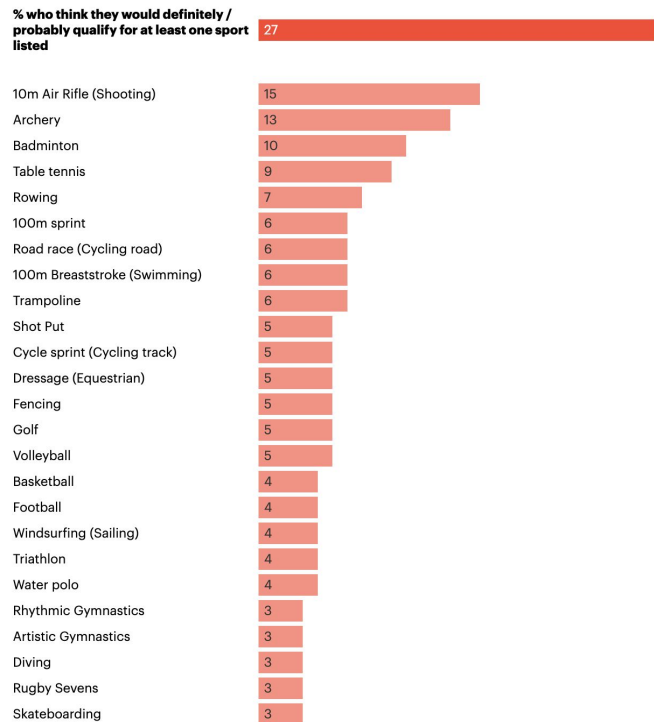
- 01. The speed at which you learn.
- 02. The new competitive advantage for your career.



Be aware that you're not self-aware.

One in four Britons reckon they could qualify for the 2028 Olympics if they started training today

The next Olympic games will be held in Los Angeles in 2028. Hypothetically speaking, if you began training TODAY, which of the following sports, if any, do you think you could qualify for the 2028 Olympics with? % saying they would definitely/probably qualify



Let's go to
Mentimeter



Verification vs. self-rating

89%

of assessments had an inaccurate self-rating

This causes persistent gaps in critical skills. People are spending a lot of time learning skills that are not critical for their roles, or that they already have.

32%

overestimation
of own proficiency level

Skills gaps would persist because people do not know they don't have the skills.

56%

underestimation
of own proficiency level

Without the skills verification from Workera, people spend time studying about skills they already have.



Tuning into expert sources for your work



Andrew Ng: [Twitter/X](#) | [LinkedIn](#)

Yann LeCun: [Twitter/X](#) | [LinkedIn](#)

Fei-Fei Li: [Twitter/X](#) | [LinkedIn](#)

François Chollet: [Twitter/X](#) | [LinkedIn](#)

Christopher Manning: [Twitter/X](#) | [LinkedIn](#)

Jensen Huang: [LinkedIn](#)

Chip Huyen: [Twitter/X](#) | [LinkedIn](#)

OpenAI: [Twitter/X](#) | [LinkedIn](#)

Anthropic: [Twitter/X](#) | [LinkedIn](#)

Google AI: [Twitter/X](#) | [LinkedIn](#)

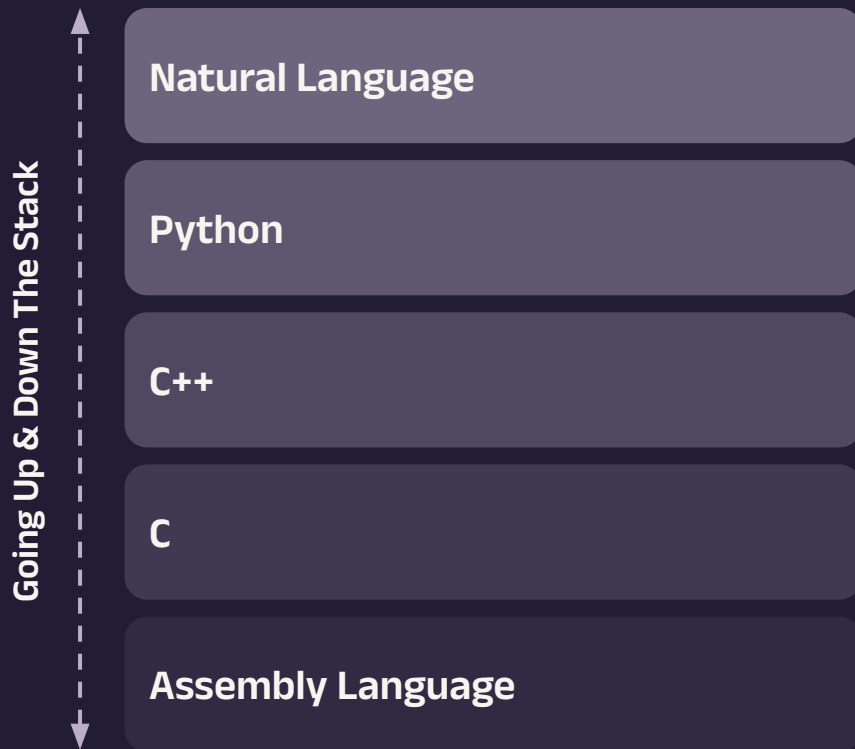
DeepLearning.AI: [LinkedIn](#)

Workera: [LinkedIn](#)

Follow



“AI is the end of coding.” Is this true?



React.js simplified significantly how we build web interfaces. It may sound counter intuitive, but it has **boosted demand for front-end engineers**.

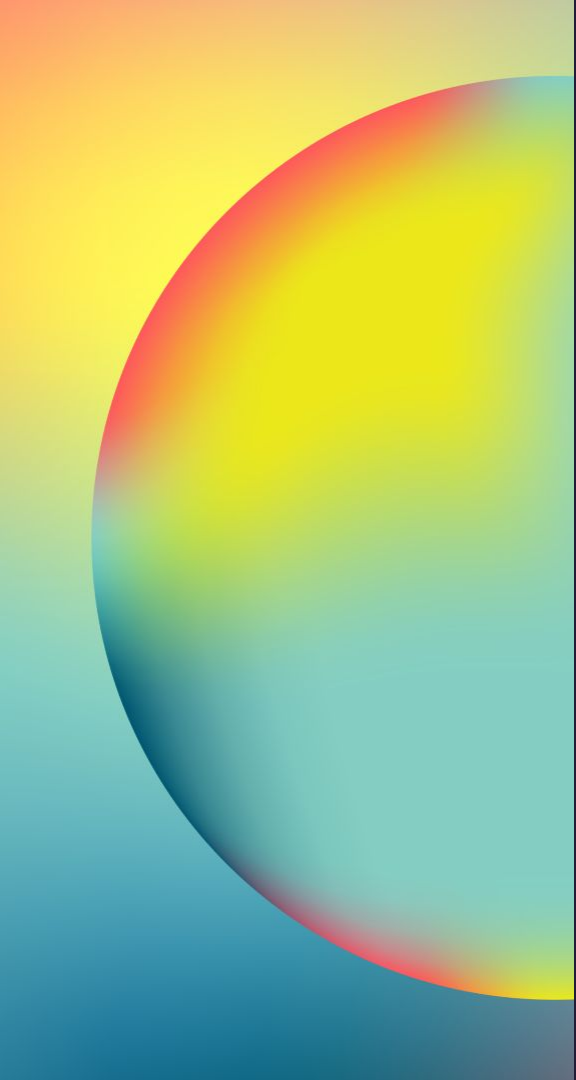
Its component-based architecture and flexibility led to **widespread adoption across industries**, creating more job opportunities.

The expanding ecosystem, including tools like Redux and React Router, further **increased the need for specialized developers**.

Closing remarks



1. **You can feel proud.** You've built a project that you can carry with you after the class.
2. If you enjoyed working on your project, **keep developing it after the class.**
3. We encourage you to **review other students' projects** which will be shared on the CS230 website.
4. **Keep in touch.** You're officially a CS230 graduate, and we hope that you've built a relationship with your team members or a teaching staff member.



Thank you